

RESEARCH ARTICLE

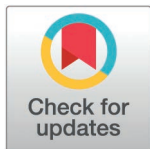
Implementation of an opioid use disorder (OUD) machine-learning phenotype in real-time for the ADAPT clinical trial

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Abstract

We developed and deploy a real-time, electronic health record-integrated machine learning phenotype to identify emergency department patients with opioid use disorder for prospective clinical trial screening and buprenorphine initiation. We conducted a multi-phase study across three emergency departments in a single United States health system from 2014 to 2025. Using visit-level data available at or before triage, we trained a random-forest classifier to estimate opioid use disorder risk and embedded scoring in the electronic health record to trigger point-of-care alerts for trial eligibility review. A high-specificity computable training label supported retrospective development, and a clinician gold-standard reference was established through structured chart review. Performance was summarized using discrimination, calibration, and threshold-based classification metrics; prospective validation used a stratified random sample of flagged and unflagged encounters. Retrospective discrimination compared to the high-specificity computable training label was high (area under the receiver operating characteristic curve, 0.99; 95% CI, 0.98–0.99; area under the precision-recall curve, 0.92; 95% CI, 0.90–0.94), and calibration plots informed operating-point selection for real-time use. In prospective gold-standard validation ($n=217$), 89.3% of model-positive encounters and 95.7% of model-negative encounters agreed with physician adjudication at the prespecified threshold. Because the sample was stratified on model prediction, these are within-stratum agreement estimates; weighting them by the source-population flag rate (28,284 of 866,569 eligible encounters, 3.26%) yields design-weighted estimates of population-level prevalence (7.1%), sensitivity (0.40, 95% CI 0.24–0.64), and specificity (0.996). An electronic health record-embedded, machine learning phenotype can accurately and feasibly identify emergency department patients with opioid use disorder in real time,

used for model development is available in [S6 Appendix](#).

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streamlining clinical trial enrollment and treatment initiation. Ongoing work will report operational metrics (e.g., alert volume and latency), monitor performance drift and equity across subgroups, and evaluate downstream clinical and trial outcomes.

Author summary

In this study, we wanted to find a better way to identify people visiting the emergency department who may be living with opioid use problems. These patients are often hard to recognize quickly, even though timely support, including offering effective medications, can greatly improve their health and safety. Traditional approaches rely heavily on clinicians noticing certain patterns in real time, which can be challenging during busy emergency care. To address this gap, we examined over a decade of information from a large health system and developed a clinical decision support tool that reviews routine information collected at the start of an emergency visit. The tool gives clinicians a simple, real-time signal when a patient may benefit from additional evaluation or treatment. Our early experience shows that this approach can help teams identify patients more consistently and connect them to care more quickly. We hope this work will make it easier for hospitals to support people affected by opioid use and to run clinical studies that improve future treatment options.

Introduction

The opioid crisis continues to evolve as one of the most urgent and complex public health emergencies in the United States. The number of overdose deaths attributed to fentanyl analogs rose by 540% from 2013 to 2016, and in 2022 alone, more than 109,000 drug overdose deaths occurred nationwide, and nearly 70% of these involved opioids [1,2]. Emergency departments (EDs) and frontline services, including emergency medical services, are frequently the first or only point of healthcare contact for individuals with opioid use disorder (OUD) at high risk of overdose. In 2018, opioids were involved in roughly one-quarter of all drug-related ED visits across the United States, and recent studies indicate that the COVID-19 pandemic further intensified opioid-related ED utilization [3,4]. Thus, EDs are critical access points for detection and intervention for OUD [3,4].

Buprenorphine is a medication for opioid use disorder (MOUD) with evidence for both its effectiveness at reducing illicit opioid use and opioid-related mortality and the viability of initiating use in the ED setting [5–8]. Despite the availability of an effective treatment and intervention, only a small subset of eligible patients, approximately 6–8%, receive buprenorphine in the ED, [9], with limitations in its prescription and uptake related to prescribers' [5–8] lack of experience with the medication, concerns for impact on workflow and drug diversion, and persisting stigma [10,11]. Our group's prior initiative, the EMBED trial, attempted to mitigate some of these gaps by developing a clinical decision support (CDS) tool for

buprenorphine initiation in the ED, but was ineffective at increasing patient-level rates of buprenorphine initiation [12]. In the ongoing ADAPT trial, we aim to build upon the work in EMBED and improve intervention effectiveness through a Multiphase Optimization Strategy Trial [13]. A crucial step toward enhancing the adoption of ED-initiated buprenorphine is the accurate identification of potentially eligible OUD patients. Automated detection of OUD within electronic health records could enable earlier intervention, facilitate clinical trial enrollment, and strengthen linkage to treatment. Patient phenotyping through machine learning offers a practical solution to real-time patient identification.

Previous studies have used machine learning to stratify individuals according to their risk of developing or currently suffering from OUD and opioid overdose based on electronic health record (EHR) data. While several models have shown a strong predictive performance in retrospective settings [14–16], a lack of workflow integration has often limited their real-world application. In parallel, Chartash et al. in the EMBED trial [17] and Taylor et al. [18] have demonstrated that EHR computable phenotypes using structured and nuanced data from patient charts can accurately identify patients with opioid-related disorders presenting to the ED. However, these phenotypes were retrospective and did not enable real-time identification of eligible patients at the point of care.

To address these gaps, we developed and implemented a real-time, EHR-integrated machine-learning OUD phenotype to identify ED patients for prospective clinical-trial screening and to support ED-initiated buprenorphine as part of a nationally disseminated, multicomponent clinical decision support (CDS) program.[13] We hypothesized that a calibrated, real-time phenotype, embedded within routine ED workflow, would achieve high positive predictive value at a prespecified threshold chosen to limit alert burden, thereby enabling timely identification of candidates for treatment and rigorous prospective evaluation.

Results

Participants and descriptive data

In the retrospective observation phase of this study, a total of 3,177,664 unique patients met the inclusion criteria, representing all adult patients with ED visits in the health system between 2014 and 2025. Following outcome-based balancing, the training dataset included 17,054 patients, and the test dataset included 4,246 patients. Within the overall population, 1,336,335 (42.05%, Table 1) were aged 18–39 years, 972,293 (30.60%) were aged 40–64 years, and 869,016 (27.35%) were 65 years or older. These age group distributions were similarly reflected in both the training and test datasets after balancing. In the overall population, 1,707,449 (53.73%) were female and 1,466,587 (46.15%) were male, with comparable distributions maintained in the training and test datasets. The racial/ethnic composition of the overall population was 59.81% White or Caucasian, 10.21% Black or African American, and 14.42% Hispanic. This composition was also similar in both the training and test datasets after balancing.

In terms of relevant past medical history (PMH), 1.57% of the overall population had a documented substance use disorder, 3.37% had mental health diagnoses, and among selected chronic illnesses, 10.21% had diabetes, 8.82% had heart disease, and 24.34% had hypertension. Regarding medication use, 1.83% had prescriptions for medications for opioid use disorder (MOUD: methadone, buprenorphine, or naltrexone), 25.73% for other opioids, and 8.34% for naloxone.

Unlike demographic characteristics, PMH and medication prevalence differed between the training and test datasets by design due to the balancing procedure. In the balanced training set, prevalence rates were higher: 9.01% for substance use disorder, 10.10% for mental health diagnoses, 13.16% for diabetes, 11.86% for heart disease, and 31.00% for hypertension. Corresponding rates in the test set were similar: 8.93%, 10.65%, 13.78%, 12.39%, and 31.44%, respectively. MOUD use was 7.83% in training and 8.83% in testing; other opioid use was 35.66% and 36.43%; naloxone med list rates were 15.39% and 16.44%.

Random forest classifier outcomes

The random forest classifier predicted the patients with a positive OUD phenotype (based on the high-specificity computable training label) with a PPV of 0.945, NPV of 0.958, sensitivity of 0.549, specificity of 0.997, FPR of 0.003, FNR of

Table 1. Study population demographics.

	Overall Population (n=3,177,664)	Balanced Training Dataset (n=17,054)	Balanced Testing Dataset (n=4,246)
Age group, n (%)			
18-39	1,336,335 (42.05%)	7,127 (41.79%)	1,761 (41.47%)
40-64	972,293 (30.60%)	5,642 (33.08%)	1,408 (33.16%)
65+	869,016 (27.35%)	4,285 (25.13%)	1,077 (25.37%)
Sex, n (%)			
Female	1,707,449 (53.73%)	8,799 (51.59%)	2,200 (52.28%)
Male	1,466,587 (46.15%)	8,246 (48.35%)	2,023 (47.64%)
Other or Unknow	3,628 (0.11%)	9 (0.05%)	3 (0.07%)
Race, n (%)			
White or Caucasian	1,900,669 (59.81%)	10,897 (63.90%)	2,788 (65.66%)
Black or African American	324,332 (10.21%)	1,931 (11.32%)	452 (10.65%)
Asian	109,581 (3.45%)	569 (3.34%)	116 (2.73%)
American Indian or Alaska Native	9,182 (0.29%)	62 (0.36%)	10 (0.24%)
Native Hawaiian or Other Pacific Islander	4,962 (0.16%)	20 (0.12%)	10 (0.24%)
Other or Unknown	828,938 (26.09%)	3,575 (20.96%)	870 (20.49%)
Ethnicity, n (%)			
Non-Hispanic	2,058,420 (64.78%)	12,461 (73.07%)	3,112 (73.29%)
Hispanic or Latino	458,068 (14.42%)	2,758 (16.17%)	682 (16.06%)
Other or Unknown	661,176 (20.81%)	1,835 (10.76%)	452 (10.64%)
PMH, n (%)>=1			
Substance Use Disorder Diagnoses	49,863 (1.57%)	1,536 (9.01%) †	379 (8.93%) †
Mental Diagnoses	107,142 (3.37%)	1,723 (10.10%)	452 (10.65%)
Chronic Illnesses			
Diabetes	324,586 (10.21%)	2,245 (13.16%)	585 (13.78%)
Heart Disease	280,286 (8.82%)	2,023 (11.86%)	526 (12.39%)
Hypertension	773,361 (24.34%)	5,286 (31.00%)	1,335 (31.44%)
Medication			
MOUD*	58,178 (1.83%)	1,336 (7.83%)	375 (8.83%)
Other opioids	817,469 (25.73%)	6,081 (35.66%)	1,547 (36.43%)
Naloxone	264,999 (8.34%)	2,624 (15.39%)	698 (16.44%)

* MOUD stands for medication of opioid use disorder and includes methadone, buprenorphine, naltrexone.

† Patients with >250 cumulative OUD ICD codes were labeled positive using a high-specificity computable training label.

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0.451, accuracy of 0.957, and F1 score of 0.694, when evaluated at the default probability threshold of 0.50. Additional evaluations using alternative probability thresholds are presented in [Table 2](#).

At the default probability threshold of 0.50, the confusion matrix demonstrated 202 true positives, 10 false positives, 3805 true negatives, and 172 false negatives ([Fig 1a](#)). The adjudicated validation sample was constructed using stratified sampling based on model predictions to ensure sufficient representation of positive and negative cases. The ROC curve yielded an AUC of 0.99 (95% CI 0.98-0.99), indicating strong overall discrimination between patients with and without OUD ([Fig 1b](#)). The precision–recall curve produced an AUC of 0.92 (95% CI 0.90 - 0.94), supporting the model's ability to

Table 2. Performance metrics of the predictive model.

Threshold	Sensitivity (TPR) [95% CI]	Specificity (TNR) [95% CI]	Precision (PPV) [95% CI]	Negative Predictive Value (NPV) [95% CI]	False Positive Rate (FPR) [95% CI]	False Negative Rate (FNR) [95% CI]	Accuracy [95% CI]	F1-Score [95% CI]
0.05	0.995 [0.987, 1.000]	0.761 [0.747, 0.775]	0.290 [0.266, 0.314]	0.999 [0.998, 1.000]	0.239 [0.225, 0.253]	0.005 [0.000, 0.013]	0.782 [0.770, 0.794]	0.449 [0.419, 0.478]
0.1	0.976 [0.960, 0.990]	0.916 [0.907, 0.924]	0.532 [0.497, 0.570]	0.997 [0.996, 0.999]	0.084 [0.076, 0.093]	0.024 [0.010, 0.040]	0.921 [0.913, 0.929]	0.689 [0.658, 0.721]
0.15	0.971 [0.953, 0.987]	0.957 [0.950, 0.963]	0.687 [0.646, 0.727]	0.997 [0.995, 0.999]	0.043 [0.037, 0.050]	0.029 [0.013, 0.047]	0.958 [0.952, 0.964]	0.804 [0.776, 0.833]
0.2	0.939 [0.914, 0.962]	0.972 [0.966, 0.977]	0.766 [0.725, 0.804]	0.994 [0.991, 0.996]	0.028 [0.023, 0.034]	0.061 [0.038, 0.086]	0.969 [0.963, 0.974]	0.844 [0.816, 0.869]
0.25	0.894 [0.859, 0.925]	0.984 [0.980, 0.988]	0.845 [0.810, 0.880]	0.990 [0.986, 0.993]	0.016 [0.012, 0.020]	0.106 [0.075, 0.141]	0.976 [0.971, 0.980]	0.869 [0.842, 0.892]
0.3	0.818 [0.776, 0.855]	0.989 [0.986, 0.992]	0.878 [0.843, 0.911]	0.982 [0.978, 0.986]	0.011 [0.008, 0.014]	0.182 [0.145, 0.224]	0.974 [0.969, 0.978]	0.847 [0.819, 0.873]
0.35	0.752 [0.710, 0.792]	0.993 [0.990, 0.995]	0.911 [0.878, 0.939]	0.976 [0.972, 0.981]	0.007 [0.005, 0.010]	0.248 [0.208, 0.290]	0.971 [0.967, 0.976]	0.824 [0.793, 0.851]
0.4	0.673 [0.624, 0.718]	0.995 [0.992, 0.997]	0.927 [0.897, 0.957]	0.969 [0.963, 0.974]	0.005 [0.003, 0.008]	0.327 [0.282, 0.376]	0.966 [0.960, 0.972]	0.780 [0.743, 0.813]
0.45	0.599 [0.548, 0.648]	0.996 [0.994, 0.998]	0.942 [0.911, 0.970]	0.962 [0.956, 0.968]	0.004 [0.002, 0.006]	0.401 [0.352, 0.452]	0.961 [0.955, 0.967]	0.732 [0.691, 0.769]
0.5	0.549 [0.502, 0.596]	0.997 [0.995, 0.998]	0.945 [0.914, 0.973]	0.958 [0.951, 0.963]	0.003 [0.002, 0.005]	0.451 [0.404, 0.498]	0.957 [0.951, 0.963]	0.694 [0.654, 0.733]
0.55	0.464 [0.415, 0.516]	0.997 [0.995, 0.998]	0.936 [0.899, 0.970]	0.950 [0.944, 0.957]	0.003 [0.002, 0.005]	0.536 [0.484, 0.585]	0.949 [0.943, 0.956]	0.621 [0.573, 0.667]
0.6	0.425 [0.376, 0.475]	0.998 [0.996, 0.999]	0.953 [0.916, 0.982]	0.947 [0.939, 0.954]	0.002 [0.001, 0.004]	0.575 [0.525, 0.624]	0.947 [0.940, 0.954]	0.588 [0.538, 0.635]
0.65	0.380 [0.334, 0.425]	0.998 [0.996, 0.999]	0.947 [0.908, 0.979]	0.943 [0.936, 0.950]	0.002 [0.001, 0.004]	0.620 [0.575, 0.666]	0.943 [0.935, 0.949]	0.542 [0.493, 0.590]
0.7	0.325 [0.278, 0.369]	0.999 [0.998, 1.000]	0.969 [0.935, 0.993]	0.938 [0.930, 0.945]	0.001 [0.000, 0.002]	0.675 [0.631, 0.722]	0.939 [0.931, 0.945]	0.486 [0.430, 0.536]
0.75	0.251 [0.207, 0.296]	0.999 [0.999, 1.000]	0.979 [0.945, 1.000]	0.932 [0.924, 0.939]	0.001 [0.000, 0.001]	0.749 [0.704, 0.793]	0.933 [0.925, 0.940]	0.399 [0.342, 0.455]
0.8	0.182 [0.144, 0.221]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.926 [0.918, 0.933]	0.000 [0.000, 0.000]	0.818 [0.779, 0.856]	0.927 [0.919, 0.934]	0.308 [0.252, 0.362]
0.85	0.108 [0.078, 0.142]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.920 [0.912, 0.928]	0.000 [0.000, 0.000]	0.892 [0.858, 0.922]	0.920 [0.913, 0.928]	0.195 [0.144, 0.248]
0.9	0.018 [0.006, 0.034]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.912 [0.904, 0.921]	0.000 [0.000, 0.000]	0.982 [0.966, 0.994]	0.912 [0.904, 0.921]	0.036 [0.011, 0.065]
0.95	0.000 [0.000, 0.000]	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	0.911 [0.902, 0.919]	0.000 [0.000, 0.000]	1.000 [1.000, 1.000]	0.911 [0.902, 0.919]	0.000 [0.000, 0.000]

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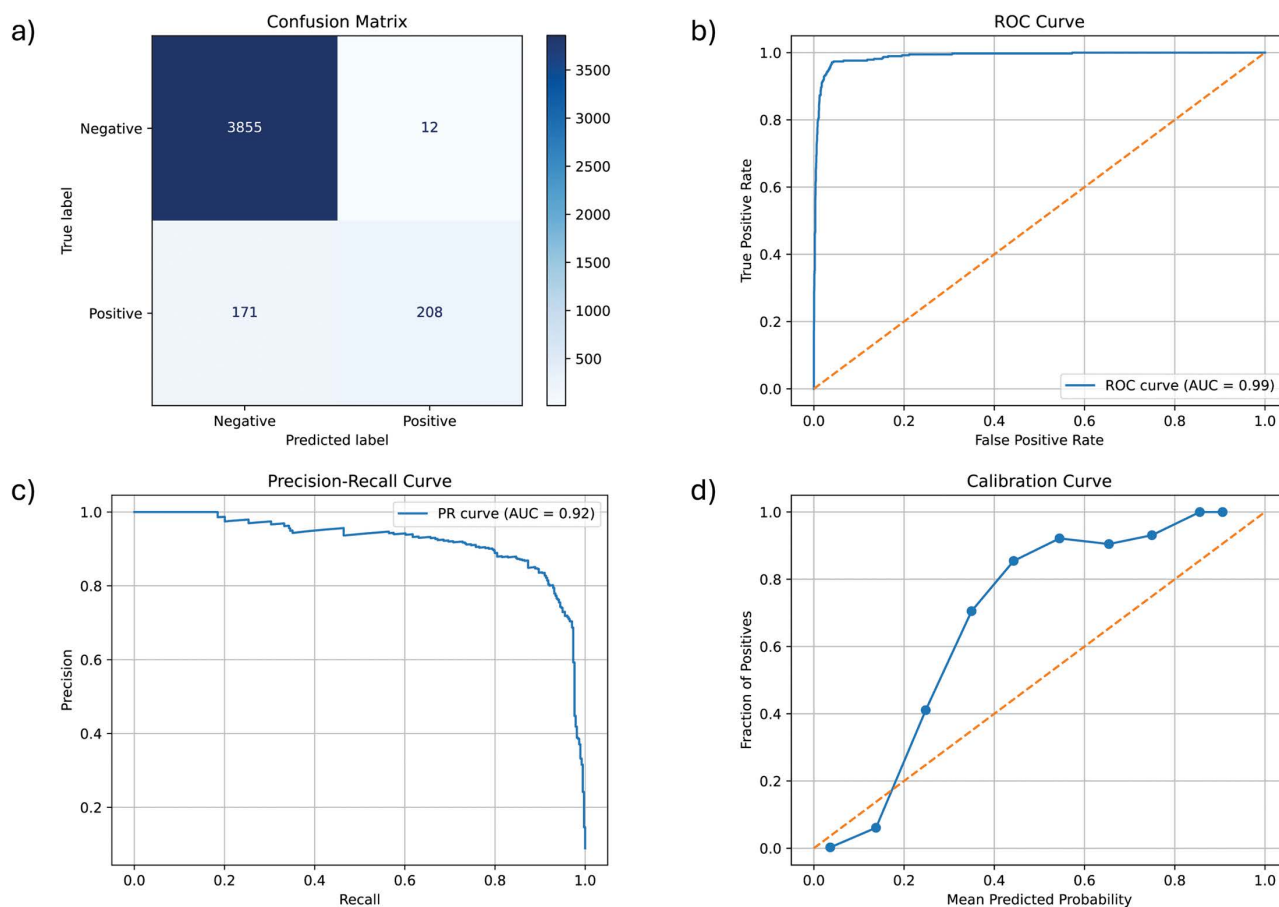


Fig 1. High-specificity computable training label retrospective prediction model performance evaluations: a) confusion matrix at the default probability threshold of 0.50, b) Receiver operating characteristic (ROC) curve with area under the curve (AUC), c) precision-recall (PR) curve with area under the curve, d) calibration curve comparing predicted versus observed probabilities.

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preserve positive predictive value under class imbalance (Fig 1c). Finally, the calibration curve demonstrated that predicted probabilities were underestimated across the observed risk spectrum, especially for higher predicted probabilities (Fig 1d). No additional recalibration was performed prior to deployment.

The top 25 predictive features, ranked by feature importance, derived from the random forest classifier, are presented in Fig 2 (full list in S1 Appendix). At the individual feature level, the most influential predictors of the OUD phenotype were diagnoses related to substance use disorders other than OUD. At the category level, features pertaining to PMH contributed the greatest overall importance, followed by medication-related variables and laboratory results. Notably, the top two of the three individual features belonged to the PMH category.

Gold standard evaluation

Across all 866,569 eligible ED encounters in 2025, 28,284 (3.26%) were flagged by the phenotype model. For the prospective validation, a stratified sample of model-positive and model-negative encounters was selected for independent physician adjudication. In the prospective validation phase of the study, there were 217 patients in the gold standard validation set, including 103 (47.47%, Table 3) classified by the model as having OUD and 114 (52.53%) classified as not

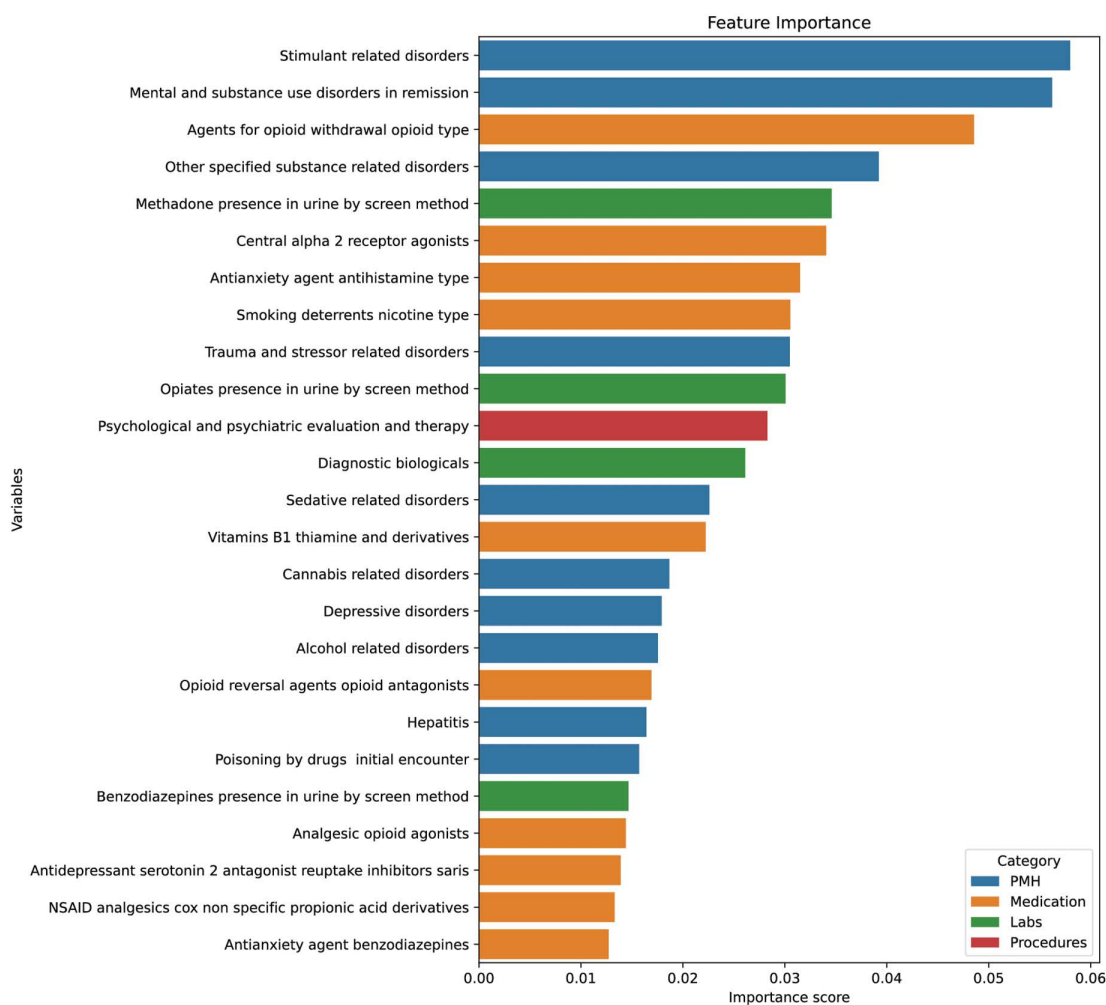


Fig 2. Feature importance of predictors for OUD phenotype: bar size represents the relative importance of each of the top 25 predictors, as estimated by the random forest classifier. Colors indicate feature categories: PMH (blue), medication (orange), labs (green), procedures (red), and other (purple).

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having OUD. We initially selected 200 patients for chart review and subsequently added 20 additional cases (for a total of 220) to ensure sufficient sample size after applying exclusion criteria. Three patients were excluded (one younger than 18 years and two who were pregnant), resulting in the final cohort size reported above. Of these patients, 42.86% (n=93) were female, and 57.14% (n=125) were male. The age distribution was as follows: 24.54% (n=53) were aged 18–39 years, 48.15% (n=105) were aged 40–64 years, and 27.31% (n=60) were aged 65 years and older. Regarding race and ethnicity, 67.28% (n=147) of patients identified as White or Caucasian, 22.58% (n=49) as Black or African American, 19.32% (n=43) as Hispanic or Latino, and 0.46% (n=1) as Asian. After chart review, 103 were ultimately adjudicated as having OUD and 114 as not having OUD. Cohen's kappa for inter-reviewer agreement was 0.86. Of the 17 cases in which resident reviewers disagreed on patients' OUD status, attending physician adjudicators initially agreed on 13 (76%) and reached a consensus on the remaining 4 cases after discussion. The demographic composition of the validation set was broadly comparable between patients with and without OUD, as determined by gold standard manual chart review. The age distribution differed slightly, with a higher proportion of patients aged 40–64 years in the OUD-positive group and a higher proportion aged 65 years and older in the OUD-negative group.

Table 3. Prospective gold standard validation group demographics.

	Overall (n = 217)	OID Yes (n = 103)	OID No (n = 114)
Age group, n (%)			
18-39	53 (24.54%)	26 (25.24%)	27 (23.89%)
40-64	104 (48.15%)	60 (58.25%)	44 (38.94%)
65+	59 (27.31%)	17 (16.50%)	42 (37.17%)
Sex, n (%)			
Female	93 (42.86%)	39 (37.86%)	54 (47.37%)
Male	124 (57.14%)	64 (62.14%)	60 (52.63%)
Unknown	0		
Race, n (%)			
White or Caucasian	146 (67.28%)	75 (72.82%)	71 (62.28%)
Black or African American	49 (22.58%)	20 (19.42%)	29 (25.44%)
Asian	1 (0.46%)	0	1 (0.88%)
American Indian or Alaska Native	0	0	0
Native Hawaiian or Other Pacific Islander	0	0	0
Other or Unknown	22 (10.14%)	8 (7.77%)	13 (11.40%)
Ethnicity, n (%)			
Non-Hispanic	170 (78.34%)	81 (78.64%)	89 (78.07%)
Hispanic or Latino	43 (19.82%)	21 (20.39%)	22 (19.30%)
Unknown	4 (1.84%)	1 (0.97%)	3 (2.63%)

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Model performance evaluated against the gold standard OUD labels in the validation set (Fig 3), within the adjudicated validation cohort, identified 92 true positives (TP), 110 true negatives (TN), 11 false positives (FP), and 5 false negatives (FN, Fig 3). Recomputed from these counts, the model yielded a PPV of 0.893 (95% CI 0.823-0.942) and an NPV of 0.957 (95% CI 0.907-0.983) with respect to the gold standard. These estimates are conditional on the stratified sampling design of the adjudicated validation cohort and should not be interpreted as population-level screening accuracy. Applying the source-population flag rate of 3.26% to the adjudicated OUD proportions within each prediction stratum yields design-weighted estimates of population-level OUD prevalence (7.1%, 95% CI 4.5-11.9), sensitivity (0.40, 95% CI 0.24-0.64), specificity (0.996, 95% CI 0.994-0.998), and expected alert burden (28,284 alerts per year across three EDs, approximately 77 per day; S2 Appendix). The interval around sensitivity is wide because it is driven by the five false negatives observed among 114 adjudicated model-negative encounters, a proportion that weighting then applies to 96.5% of the source population.

Discussion

In this multi-phase study, we developed, implemented, and prospectively validated an EHR-embedded phenotype that accurately identifies patients with OUD in real time. By integrating daily inference within existing data infrastructure, we demonstrated that automated phenotyping can transition from retrospective research to prospective clinical application with minimal latency. The model's high predictive performance and operational feasibility highlight its potential to enhance clinical care and clinical trial efficiency for ED patients with OUD.

The phenotype classifier demonstrated excellent retrospective performance in identifying patients with OUD, achieving a high AUC of 0.99 in the high-specificity computable training label evaluation based on diagnostic codes, and showed concordance with independent physician adjudication within the stratified validation sample, where 92 of 103 model-positive encounters met DSM-5 criteria for OUD. Retrospective discrimination was measured against a computable training label rather than a clinical reference standard, so it is not directly comparable to prior EHR-based models for substance use disorders validated against chart review, which have reported values from 0.80 to 0.95 [19,20]. Several

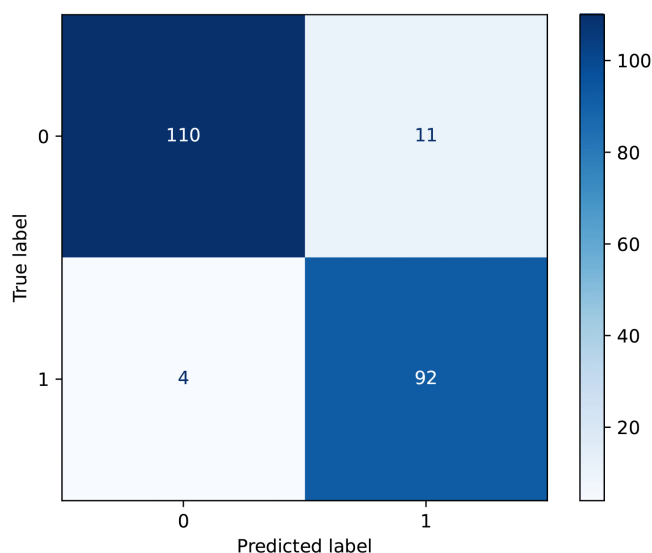


Fig 3. Prospective gold standard validation outcome confusion metrics. The figure presents the distribution of TP, FP, TN, and FN from the validation set. Each cell represents the count of patients correctly or incorrectly classified by the model after chart review based on the gold standard. The model demonstrated high agreement with the gold standard.

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factors may have contributed to this strong performance, including the use of a high-specificity computable training label, the construction of features from detailed EHR data that captured robust patterns indicative of OUD, and the application of tree-based models, which are well-suited for handling non-linear relationships and complex interactions among clinical features. Additionally, the model's predictions were highly consistent with the gold-standard chart review outcomes, demonstrating strong alignment between algorithmic classification and clinical validation. The prevalence of OUD differed across study phases. During model training, majority-class undersampling was applied to address class imbalance and improve model learning for the minority class. For model development with respect to the high-specificity computable training label, the dataset was first balanced using majority class undersampling followed by minority class oversampling, and the resulting dataset was then split into training and test sets. The gold standard adjudicated validation sample was constructed using stratified sampling based on model predictions and therefore does not reflect natural prevalence; instead, it allows us to report on positive and negative predictive of model predictions in our ED population at a prespecified threshold. Because positive and negative predictive values depend on both the classification threshold and the prevalence of the outcome in the target population, these values should be recalculated when applying the model to different populations or clinical settings.

The pipeline was designed to operate within standard EHR data structures and executed efficiently using distributed computing, demonstrating technical feasibility for real-time integration. Our implementation also differs from prior work by closing the loop operationally—through nightly OMOP harmonization, daily inference, and EHR-native rules—yielding ≤ 24 -hour latency [21]. These findings suggest that real-time deployment within the EHR environment is both technically feasible and clinically effective, enabling timely identification of patients at risk for OUD who could benefit from appropriate ED intervention and treatment. While the EMBED pragmatic trial demonstrated that CDS alone does not guarantee population-level uptake [22]; the new phenotype presented here could address these challenges by offering a maintainable pathway for automated, real-time screening to support increased patient identification and subsequent treatment. Finally, multi-site rule sets have demonstrated portability across health systems; by grounding our pipeline in standard vocabularies and OMOP, we similarly position the phenotype for reuse and external validation [17].

Prior ED phenotyping efforts have primarily relied on rule-based algorithms anchored to diagnosis codes and a small set of structured signals. While precise, these approaches: 1) under-detect uncoded OUD, and 2) do not allow for detection and intervention during the encounter itself (given diagnostic codes are typically applied at the end of or after the encounter [14,17,23,24]. More recent machine-learning models for OUD and overdose risk report strong retrospective discrimination, but most remain decoupled from real-time, clinical workflow, with limited attention to calibration, latency, or operational usability [18]. In contrast, our strategy paired a high-specificity computable training label for model development with DSM-5-aligned gold-standard adjudication for prospective evaluation. This approach reduced label noise by limiting misclassified or ambiguous cases in the training data, while quantifying real-world performance. Unlike diagnostic code-only phenotypes, we used count-based longitudinal features (diagnoses, medications, laboratory positives) to preserve burden and utilization signal, which may explain the higher discrimination observed. Finally, few prior studies report fairness or drift monitoring plans [14,17]; we explicitly outline future equity audits and recalibrations, which are essential for sustained, trustworthy deployment in dynamic ED environments. These audits will examine model performance across patient demographic groups and under varying patterns of data availability, with particular attention to variables that may reflect structural barriers to care access or clinician decision-making patterns. Examples include substance use disorder diagnostic codes [25], prior receipt of medications for opioid use disorder or opioid reversal agents [25–27], urine drug screening, and health care visit intensity [28].

Our findings demonstrate that an electronic phenotype can reliably identify patients with OUD in the ED in real time, offering a practical bridge between predictive modeling and actionable clinical use. Such integration has immediate implications for both research and care delivery: it can streamline enrollment in clinical trials such as the ADAPT Trial [13], thereby reducing the manual screening burden and ensuring that eligible patients are identified and offered evidence-based treatments such as buprenorphine. Beyond trial operations, real-time OUD detection can facilitate more consistent linkage to addiction medicine and recovery resources, reducing variability across clinicians and encounters. As health systems expand the use of phenotyping pipelines, this approach could serve as a generalizable model for identifying other high-risk conditions requiring timely intervention in the ED and beyond.

This study has several limitations. First, it was conducted within a single academic health system using a single EHR vendor product and locally harmonized data; results may not generalize to other institutions, vendors, or patient populations. Second, the retrospective high-specificity computable training label, based on high counts of OUD-related ICD codes, may misclassify patients due to documentation or utilization bias. Because both the training label and the count-based features encode cumulative health-system contact, retrospective discrimination partly reflects documentation density rather than OUD alone, and patients with true OUD but limited prior contact are the group most likely to be labeled and predicted negative. Additionally, the training label was designed as a high-specificity training anchor rather than a clinical definition of OUD. As such, it prioritized reducing label noise over capturing the full clinical spectrum of OUD. Third, the prospective validation used stratified sampling based on model predictions rather than natural prevalence, so unweighted sensitivity and specificity do not apply to the ED population and are reported only as design-weighted estimates that depend on the source-population flag rate. Fourth, the model relied on structured EHR data, omitting the potentially informative narrative content of clinical notes, and used a simple imputation method (global mode) that may introduce bias by over-representing common values, obscuring true variability across patients or sites, and failing to account for potentially informative missingness in ED EHR data; undersampling and within-system validation may also overestimate performance. Fifth, although demographic variables were excluded from modeling, bias may persist through proxy features. Sixth, because clinician-entered EHR informational records (problem lists and diagnosis codes) are variably incomplete and can be insensitive to substance-use conditions (problem-list completeness 60.2–99.4% across 10 sites; ICD code algorithms for substance use disorder sensitivity 47–83% vs medical-record validation) [29,30], we used a high-cardinality, inclusive set of OUD-related diagnosis codes to improve sensitivity and reduce under-ascertainment, although residual documentation and coding variation may still introduce misclassification [31]. Lastly, the prospective

validation cohort was created using stratified sampling of model-positive and model-negative encounters to ensure adequate representation of both prediction strata. As a result, this phase was designed to evaluate model performance within the sampled prediction strata rather than estimate population-level screening performance.

Future work should focus on evaluating the clinical and operational impact of real-time OUD phenotyping in prospective, multi-site trials. Specifically, studies should assess whether automated identification improves rates of ED-initiated buprenorphine, treatment linkage, and long-term recovery outcomes. Ongoing monitoring for performance drift, calibration decay, and subgroup disparities will be critical to maintaining accuracy and fairness over time. Finally, extending this framework to other conditions, such as alcohol use disorder, suicidality, or sepsis in the ED (or beyond) could demonstrate the broader potential of embedded phenotyping to enhance real-time clinical decision support.

Methods

Study design and study population

This multi-phase study involved two components: 1) a retrospective observational study utilizing ED visit data from 2014 to 2025 to develop a predictive model, and 2) a prospective validation of the resulting model. The validated model was created with the goal of integration into a clinical decision support tool designed to identify OUD patients eligible for buprenorphine initiation in the prospective ADAPT clinical trial [13]. This manuscript describes model development and prospective phenotype validation in preparation for the ADAPT trial; results of the trial itself will be reported after trial completion.

The model was trained on patient data from across the regional health system, including nine EDs. The study population for this phase included all adult patients aged 18 and older who had not opted out of research use. This study employed the STROBE reporting guidelines for observational research [32]. All data processing, feature engineering, and model development were performed using PySpark (*PySpark v3.2.1*) on our health system's local cloud-based secure computing platform. The descriptive statistics for the study population, including the overall cohort, the train/test subsets (after balancing and splitting), are summarized in [Table 1](#). Phenotype development was deemed exempt from human subject review by our institution's IRB (protocol #2000037541), and prospective use in the ADAPT clinical trial was approved (protocol # 2000038624) with a waiver of informed consent. The ADAPT clinical trial is registered on clinicaltrials.gov (ID: NCT06799117).

Data source, variables, and measurements

All patient data were extracted from the health system's Epic EHR (Epic, Verona, WI) data systems (Clarity and Caboodle, [Fig 4](#), pipeline diagram). Using established extract–transform–load (ETL) procedures, records were ingested into the local Computational Health Platform (CHP) and harmonized to the OMOP common data model for analysis. Routine data quality checks (completeness, conformance, and plausibility) were applied at each stage of the pipeline. Demographic information, including age, gender, race, and ethnicity, was collected but not included as model features. Past medical history items were categorized according to Agency for Healthcare Research and Quality (AHRQ) Clinical Classification Software (CSS) categories [33], along with an additional variable that counted instances of OUD based on the provided ICD code value set ([S3 Appendix](#)) [34]. Medication exposures and orders were summarized as counts rather than simple presence/absence, and for pre-specified laboratory tests, we recorded the number of positive results. All feature tables were joined on the unique patient identifier. Then, variable names were normalized (lowercase; special characters removed), and missing values were imputed using the variable-wise mode. This count-based representation was selected because it preserves longitudinal burden and utilization signal compared with binary indicators.

High-specificity computable training label. Since OUD prevalence in our ED cohort is low and manual physician adjudication is resource-intensive, we trained the model using a high-specificity computable training label rather than

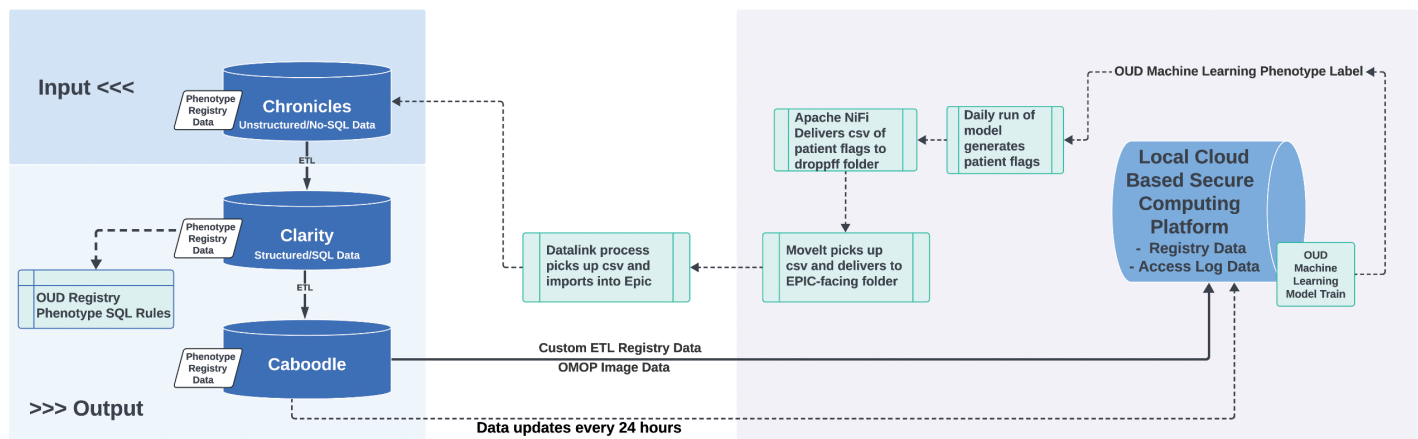


Fig 4. System architecture and data flow: this pipeline diagram illustrates the multi-phase process of data abstraction, model development, and deployment of a pre-trained OUD prediction model. It displays how the model was integrated into the EHR system to support real-time CDS for the timely identification of eligible ED patients. Chronicles is Epic's foundational operational database that stores real-time patient-level clinical data. Clarity is a relational reporting database that extracts and standardizes data from Chronicles for analytical use. Carboodles is an analytics-ready data warehouse built on top of both Chronicles and Clarity, designed to support research and operational reporting. The local cloud-based secure computing platform serves as the environment in which the model is built, validated, and operated, using the OMOP common data model, which is a Common Data Model that YNH database engineers extract from the Epic Caboodle database according to OHDSI specifications. Apache NiFi, deployed via Docker, facilitates automated and reproducible data ingestion and transformation workflows across systems. Within Epic, Datalink was utilized to enable targeted queries and cohort selection directly from the EHR.

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a physician-adjudicated gold standard. The outcome was defined from longitudinal EHR data as the cumulative count of OUD-related ICD codes (value set in [S3 Appendix](#)) recorded in each patient's medical record. Patients with >250 cumulative OUD codes were labeled positive for OUD. This high threshold was selected to avoid false positives from high-frequency copy-and-paste documentation habits as well as diagnosis codes existing in multiple places in the chart (such as problem list, past medical history, and encounter diagnoses), resulting in a bloated number of diagnosis codes and refined through targeted clinical chart reviews to prioritize specificity and maximize positive predictive value (PPV) [35], while also optimizing sensitivity ([S4 Appendix](#)). For downstream flagging, we created a composite operational label to improve sensitivity: a patient was considered positive if either (i) the model predicted OUD or (ii) the cumulative OUD ICD count was ≥ 250 . This rule preserved the high-specificity anchor used for training while broadening capture to patients with substantial—but less extreme—documentation density.

Gold standard. We used physician chart review consistent with the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) definition of OUD as our gold standard [36], with full methods detailed provided in [S5 Appendix](#).

Model development and validation

Phenotyping prediction algorithm. A random forest classifier was utilized to predict patient OUD status at the start of an ED encounter based on the high-specificity computable training label definition (XGBoost was not available in our PySpark, secure compute environment; we also evaluated logistic regression (with and without L1/LASSO)), but Random Forest delivered higher AUROC/PR-AUC and comparable calibration on cross-validation, so it was selected as the primary model). Tree-based models are well known for their strong generalization performance, high predictive accuracy, and interpretability [37]. To address the imbalance in the dataset, we applied undersampling to the majority class (non-OUD) through random sampling based on the defined outcomes, thereby aligning its frequency with that of

the positive class (OUD), and achieved a 10:1 ratio. Subsequently, we performed feature engineering using PySpark's VectorAssembler to consolidate relevant features into a single vector column. The balanced dataset was then divided into training and testing subsets, with an 80:20 ratio, consistent with standard practice in model development. The random forest classifier was trained on the training data, with hyperparameters optimized using a Binary Classification Evaluator [38,39]. This evaluator assessed model performance based on the area under the ROC curve (AUC), and 2-fold cross-validation was performed to mitigate the risk of overfitting. Code used for model development and evaluation is available in [S6 Appendix](#).

Evaluation analysis

Phase 1 High-specificity computable training label evaluation. Model performance compared to the high-specificity computable training label was evaluated using confusion matrices (i.e., 2×2 contingency tables, [Fig 1a](#)), along with additional metrics including sensitivity (recall), specificity, PPV (precision, the proportion of predicted positives that were true positives), negative predictive value (NPV), false positive rate (FPR), false negative rate (FNR), accuracy, and F1 score (harmonic mean of precision and recall, [Table 2](#)). The ROC AUC and the area under the precision-recall curve (PR AUC) were also calculated and plotted ([Fig 1b](#), [1c](#)). In addition, a calibration curve ([Fig 1d](#)) was plotted to assess the model's goodness of fit. Feature importance was also generated and plotted to better understand the association between features and the outcome variable. Feature importance was calculated from the trained random forest model using the Gini impurity-based importance measure, which quantifies the relative contribution of each feature to model prediction. Features were ranked from highest to lowest importance based on this measure.

Phase 2 Gold standard evaluation: Chart review. To prospectively evaluate model predictions through stratified clinician adjudication, four resident physicians (SF, JM, CRK, OR) with extensive experience caring for OUD patients conducted manual chart review of randomly-selected cases. We selected cases based on model classification to address class imbalance (patients classified as not having OUD far outnumbered those classified as having OUD), randomly selecting 110 patients from each prediction class (OUD positive or negative by model prediction), with the goal of retaining at least 100 patients in each class and a total of 200 patients after exclusions. This stratified sampling approach was intended to ensure adequate representation of both prediction strata for physician adjudication and was not designed to estimate population-level screening performance. Sampling was restricted to a single visit per patient. To ensure consistency with the Epic CDS alert algorithm logic and the inclusion criteria used in prior phases, we applied exclusion criteria to remove cases involving pregnant patients or patients under 18 years of age.

Each case was reviewed by two resident physicians, who were given a medical record number and date of encounter (the date at which the model prediction was made) and asked to determine whether the patient met criteria for a diagnosis of OUD according to the DSM-5 definition at the start of the encounter; reviewers were blinded to the model's predictions. To ensure consistency and accuracy in labeling, reviewers followed structured chart review guidance, adapted from Pulumbo et al. [40] as detailed in [S5 Appendix](#). In addition to structured data elements, reviewers were directed to review unstructured clinical notes, which could contain data and details not included in predictive model features, all of which were based on structured data elements. If reviewers noted that they had cared for a patient whose chart they were reviewing, the patient was excluded to avoid bias related to personal clinical experience.

Reviewers' assessments of each patient's OUD status were compared, and inter-reviewer agreement was assessed with Cohen's kappa statistic. In cases in which both reviewers agreed on the patient's OUD status, their assessment was considered the gold standard. In cases of disagreement, two attending physicians (EM and MI) with expertise in OUD reviewed the cases independently; if they both agreed, their assessment was considered the gold standard. If they disagreed, they discussed the case to reach a final consensus gold-standard diagnosis.

Model performance with respect to the gold standard was reported as positive predictive value (PPV, the proportion of phenotype-positive cases that met gold-standard criteria for OUD) and negative predictive value (NPV, the proportion of

phenotype-negative cases that did not meet gold standard criteria for OUD). These metrics were chosen to directly assess the clinical relevance of the phenotype in identifying true positive and negative cases of OUD, particularly in the context of DSM-5 criteria [17]. Because the validation cohort was generated through stratified sampling of model-positive and model-negative encounters, these metrics describe agreement within the sampled prediction strata and were not intended to estimate population-level screening performance.

Real-time implementation

In our real-time implementation of this machine-learning OUD phenotype, on a nightly (~24-hour) schedule, encounters and longitudinal EHR elements were extracted from Epic Clarity/Caboodle into the institutional Computational Health Platform (CHP) and harmonized to the OMOP common data model. The trained OUD random forest classifier was then executed as a daily batch job across the OMOP tables to produce a patient-level phenotype probability and binary flag for all eligible patients. Flags were published back to Epic via Apache NiFi and Epic DataLink as CSV payloads to a monitored drop location and ingested into production, yielding ≤24-hour latency between documentation and flag availability (Fig 4).

At ED arrival/registration, the real-time phenotype rules implemented in the Epic EHR consumed the latest model flag and applied additional clinical logic. Adults (≥18 years) triggered an alert if they either met the model's operating threshold (selected during validation to balance PPV and sensitivity) or presented with OUD-relevant chief complaints, received pre-hospital naloxone, or had triage documentation of a clinical opiate withdrawal scale (COWS) score >7. Alerts were suppressed for patients with an existing prescription in the predefined medication grouping (given the trial's goal to initiate medication treatment for OUD, S7 Appendix), pregnancy, or documented research opt-out.

Operating-point governance

The primary deployment objective was to provide a high-confidence signal for ADAPT screening and ED-initiated buprenorphine while maintaining an acceptable alert burden. Given finite clinical/research staffing for real-time eligibility review and the well-described risk of alert fatigue in the ED, threshold selection prioritized specificity/PPV (i.e., minimizing false positives and unnecessary interruptions) over maximal sensitivity. The operating point was selected using calibration-informed evaluation and prespecified workflow considerations (e.g., expected alerts per shift/site), with the intent that each alert would be sufficiently actionable to justify clinician interruption and trial-screening effort. We clarified that sensitivity is supported operationally through complementary rule logic already described in your implementation section (e.g., OUD-relevant chief complaint, prehospital naloxone, COWS >7), which acts as a "safety net" for patients who may not meet the high-specificity model threshold at the time of flag consumption.

Supporting information

S1 Appendix. Top 50 feature importance from random forest classifier.

(DOCX)

S2 Appendix. Design-weighted population-level estimates from the stratified prospective validation and code used for their calculation.

(DOCX)

S3 Appendix. Opioid use disorder ICD code value set.

(DOCX)

S4 Appendix. Summary statistics for opioid use disorder counts per patient per day.

(DOCX)

S5 Appendix. Chart review guidelines.

(DOCX)

S6 Appendix. Code used for phenotype model development and evaluation.

(PY)

S7 Appendix. Prescribed medications list.

(DOCX)

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Disclaimer

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References

1. Tanz LJ, Gladden RM, Dinwiddie AT. Routes of drug use among drug overdose deaths — United States, 2020–2022. *MMWR Morb Mortal Wkly Rep*. 2024;73:124–30. <http://dx.doi.org/10.15585/mmwr.mm7306a2>
2. Boté SH. U.S. opioid epidemic: impact on public health and review of Prescription Drug Monitoring Programs (PDMPs). *Online J Public Health Inform*. 2019;11(2):e18. <https://doi.org/10.5210/ojphi.v11i2.10113> PMID: [31632612](https://pubmed.ncbi.nlm.nih.gov/31632612/)
3. Pickens CM, Hoots BE, Casillas SM, Scholl L. Prevalences of and characteristics associated with single- and polydrug-involved U.S. emergency department visits in 2018. *Addict Behav*. 2022;125:107158.
4. Moghtaderi A, Zocchi MS, Pines JM, Venkat A, Black B. Estimating the uncertain effect of the COVID pandemic on drug overdoses. *PLoS One*. 2023;18(8):e0281227. <https://doi.org/10.1371/journal.pone.0281227> PMID: [37561686](https://pubmed.ncbi.nlm.nih.gov/37561686/)
5. Wakeman SE, Larochelle MR, Ameli O. Comparative effectiveness of different treatment pathways for opioid use disorder. *JAMA Netw Open*. 2020;7(5). <https://doi.org/10.1001/jamanetworkopen.2024.19798>

6. D'Onofrio G, O'Connor PG, Pantalon MV, Chawarski MC, Busch SH, Owens PH, et al. Emergency department-initiated buprenorphine/naloxone treatment for opioid dependence: a randomized clinical trial. *JAMA*. 2015;313(16):1636–44. <https://doi.org/10.1001/jama.2015.3474> PMID: [25919527](https://pubmed.ncbi.nlm.nih.gov/25919527/)
7. Herring AA, Perrone J, Nelson LS. Managing opioid withdrawal in the emergency department with buprenorphine. *Ann Emerg Med*. 2019;73(5):481–7. <https://doi.org/10.1016/j.annemergmed.2018.11.032> PMID: [30616926](https://pubmed.ncbi.nlm.nih.gov/30616926/)
8. Hawk K, Hoppe J, Ketcham E, LaPietra A, Moulin A, Nelson L, et al. Consensus recommendations on the treatment of opioid use disorder in the emergency department. *Ann Emerg Med*. 2021;78(3):434–42. <https://doi.org/10.1016/j.annemergmed.2021.04.023> PMID: [34172303](https://pubmed.ncbi.nlm.nih.gov/34172303/)
9. Chhabra N, Smith D, Dickinson G. Trends and disparities in initiation of buprenorphine in US emergency departments, 2013–2022. *JAMA Netw Open*. 2024;7(9):e2435603.
10. Cao SS, Dunham SI, Simpson SA. Prescribing buprenorphine for opioid use disorders in the ED: a review of best practices, barriers, and future directions. *Open Access Emerg Med*. 2020;12:261–74. <https://doi.org/10.2147/OAEM.S267416> PMID: [33116962](https://pubmed.ncbi.nlm.nih.gov/33116962/)
11. Hawk KF, D'Onofrio G, Chawarski MC, O'Connor PG, Cowan E, Lyons MS, et al. Barriers and Facilitators to clinician readiness to provide emergency department-initiated buprenorphine. *JAMA Netw Open*. 2020;3(5):e204561. <https://doi.org/10.1001/jamanetworkopen.2020.4561> PMID: [32391893](https://pubmed.ncbi.nlm.nih.gov/32391893/)
12. Melnick ER, Nath B, Dziura JD, Casey MF, Jeffery MM, Paek H, et al. User centered clinical decision support to implement initiation of buprenorphine for opioid use disorder in the emergency department: EMBED pragmatic cluster randomized controlled trial. *BMJ*. 2022;377:e069271. <https://doi.org/10.1136/bmj-2021-069271> PMID: [35760423](https://pubmed.ncbi.nlm.nih.gov/35760423/)
13. Iscoe MS, Diniz Hooper C, Levy DR, Buchanan L, Dziura J, Meeker D, et al. Adaptive decision support for addiction treatment to implement initiation of buprenorphine for opioid use disorder in the emergency department: protocol for the ADAPT Multiphase Optimization Strategy trial. *BMJ Open*. 2025;15(2):e098072. <https://doi.org/10.1136/bmjopen-2024-098072> PMID: [39979056](https://pubmed.ncbi.nlm.nih.gov/39979056/)
14. Ranapurwala SI, Alam IZ, Pence BW, Carey TS, Christensen S, Clark M, et al. Development and validation of an electronic health records-based opioid use disorder algorithm by expert clinical adjudication among patients with prescribed opioids. *Pharmacoepidemiol Drug Saf*. 2023;32(5):577–85. <https://doi.org/10.1002/pds.5591> PMID: [36585827](https://pubmed.ncbi.nlm.nih.gov/36585827/)
15. Egualé T, Bastardot F, Song W, Motta-Calderson D, Elsobky Y, Rui A, et al. A machine learning application to classify patients at differing levels of risk of opioid use disorder: clinician-based validation study. *JMIR Med Inform*. 2024;12:e53625. <https://doi.org/10.2196/53625> PMID: [38842167](https://pubmed.ncbi.nlm.nih.gov/38842167/)
16. Lo-Ciganic W-H, Huang JL, Zhang HH, Weiss JC, Wu Y, Kwok CK, et al. Evaluation of machine-learning algorithms for predicting opioid overdose risk among medicare beneficiaries with opioid prescriptions. *JAMA Netw Open*. 2019;2(3):e190968. <https://doi.org/10.1001/jamanetworkopen.2019.0968> PMID: [30901048](https://pubmed.ncbi.nlm.nih.gov/30901048/)
17. Chartash D, Paek H, Dziura JD, Ross BK, Nogee DP, Boccio E, et al. Identifying opioid use disorder in the emergency department: multi-system electronic health record-based computable phenotype derivation and validation study. *JMIR Med Inform*. 2019;7(4):e15794. <https://doi.org/10.2196/15794> PMID: [31674913](https://pubmed.ncbi.nlm.nih.gov/31674913/)
18. Taylor RA, Gilson A, Schulz W, Lopez K, Young P, Pandya S, et al. Computational phenotypes for patients with opioid-related disorders presenting to the emergency department. *PLoS One*. 2023;18(9):e0291572. <https://doi.org/10.1371/journal.pone.0291572> PMID: [37713393](https://pubmed.ncbi.nlm.nih.gov/37713393/)
19. Banks TJ, Nguyen TD, Uhlmann JK, Nair SS, Scherrer JF. Predicting opioid use disorder before and after the opioid prescribing peak in the United States: a machine learning tool using electronic healthcare records. *Health Informatics J*. 2023;29(2):14604582231168826. <https://doi.org/10.1177/14604582231168826> PMID: [37042333](https://pubmed.ncbi.nlm.nih.gov/37042333/)
20. Song SL, Dandapani HG, Estrada RS, Jones NW, Samuels EA, Ranney ML. Predictive models to assess risk of persistent opioid use, opioid use disorder, and overdose. *J Addict Med*. 2024;18(3):218–39. <https://doi.org/10.1097/ADM.0000000000001276> PMID: [38591783](https://pubmed.ncbi.nlm.nih.gov/38591783/)
21. Poulsen MN, Freda PJ, Troiani V, Davoudi A, Mowery DL. Classifying characteristics of opioid use disorder from hospital discharge summaries using natural language processing. *Front Public Health*. 2022;10:850619. <https://doi.org/10.3389/fpubh.2022.850619> PMID: [35615042](https://pubmed.ncbi.nlm.nih.gov/35615042/)
22. Mccamic E. EMBED publishes results from study of buprenorphine initiation in emergency departments. Available from: <https://rethinkingclinicaltrials.org/news/june-27-2022-embed-trial-did-not-increase-rate-of-emergency-buprenorphine-initiation-for-opioid-use-disorder-increased-number-of-physicians-using-the-lifesaving-practice/>. 2022. Accessed 2025 October 8.
23. Poulsen MN, Nordberg CM, Troiani V, Berrettini W, Asdell PB, Schwartz BS. Identification of opioid use disorder using electronic health records: beyond diagnostic codes. *Drug Alcohol Depend*. 2023;251:110950. <https://doi.org/10.1016/j.drugalcdep.2023.110950> PMID: [37716289](https://pubmed.ncbi.nlm.nih.gov/37716289/)
24. Venkatesh A, Malicki C, Hawk K, D'Onofrio G, Kinsman J, Taylor A. Assessing the readiness of digital data infrastructure for opioid use disorder research. *Addict Sci Clin Pract*. 2020;15(1):24. <https://doi.org/10.1186/s13722-020-00198-3> PMID: [32650817](https://pubmed.ncbi.nlm.nih.gov/32650817/)
25. Williams EC, Fletcher OV, Frost MC, Harris AHS, Washington DL, Hoggatt KJ. Comparison of substance use disorder diagnosis rates from electronic health record data with substance use disorder prevalence rates reported in surveys across sociodemographic groups in the veterans health administration. *JAMA Netw Open*. 2022;5(6):e2219651. <https://doi.org/10.1001/jamanetworkopen.2022.19651> PMID: [35771574](https://pubmed.ncbi.nlm.nih.gov/35771574/)
26. Chieh KY, Walter LA, Cropsey KL, Li L. Rates of buprenorphine prescribing and racial disparities among patients with opioid overdose. *Drug Alcohol Depend Rep*. 2024;13:100298. <https://doi.org/10.1016/j.dadr.2024.100298> PMID: [39583304](https://pubmed.ncbi.nlm.nih.gov/39583304/)
27. Barnett ML, Meara E, Lewinson T, Hardy B, Chyn D, Onsando M, et al. Racial inequality in receipt of medications for opioid use disorder. *N Engl J Med*. 2023;388(19):1779–89. <https://doi.org/10.1056/NEJMsa2212412> PMID: [37163624](https://pubmed.ncbi.nlm.nih.gov/37163624/)

28. Ortiz-Guerrero G, Logan M, Uysal U, Landazuri P, Jackson L, Ulloa CM. Racial disparities in urine drug screening among seizure patients in the emergency department. *PLoS One*. 2025;20(9):e0331908. <https://doi.org/10.1371/journal.pone.0331908> PMID: [40929160](#)
29. Wright A, McCoy AB, Hickman T-TT, Hilaire DS, Borbolla D, Bowes WA 3rd, et al. Problem list completeness in electronic health records: a multi-site study and assessment of success factors. *Int J Med Inform*. 2015;84(10):784–90. <https://doi.org/10.1016/j.ijmedinf.2015.06.011> PMID: [26228650](#)
30. McGrew KM, Homco JB, Garwe T, Dao HD, Williams MB, Drevets DA, et al. Validity of International Classification of Diseases codes in identifying illicit drug use target conditions using medical record data as a reference standard: a systematic review. *Drug Alcohol Depend*. 2020;208:107825. <https://doi.org/10.1016/j.drugalcdep.2019.107825> PMID: [31982637](#)
31. Weiskopf NG, Weng C. Methods and dimensions of electronic health record data quality assessment: enabling reuse for clinical research. *J Am Med Inform Assoc*. 2013;20(1):144–51. <https://doi.org/10.1136/amiajnl-2011-000681> PMID: [22733976](#)
32. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) Statement: Guidelines for Reporting Observational Studies. Available from: <https://www.equator-network.org/reporting-guidelines/strobe/>. Accessed 2025 October 8.
33. Clinical Classifications Software (CCS) for ICD-9-CM. Available from: <https://hcup-us.ahrq.gov/toolssoftware/ccs/ccs.jsp>. Accessed 2025 September 11.
34. Website. Available from: <https://www.acep.org/cedr/measures>
35. Holmgren AJ, Downing NL, Bates DW, Shanafelt TD, Milstein A, Sharp CD, et al. Assessment of electronic health record use between US and non-US health systems. *JAMA Intern Med*. 2021;181(2):251–9. <https://doi.org/10.1001/jamainternmed.2020.7071> PMID: [33315048](#)
36. American Psychiatric Association. Diagnostic and statistical manual of mental disorders; 2022.
37. Sullivan W. Decision tree and random forest: machine learning and algorithms: the future is here!. Createspace Independent Publishing Platform; 2018.
38. Ayan D. Evaluating binary classification models with PySpark. Medium. Available from: <https://medium.com/@demrahayan/evaluating-binary-classification-models-with-pyspark-2afc5ac7937f>; 2024. Accessed 2025 July 7.
39. BinaryClassificationEvaluator — PySpark 4.0.0 documentation. Available from: <https://spark.apache.org/docs/latest/api/python/reference/api/pyspark.ml.evaluation.BinaryClassificationEvaluator.html>. Accessed 2025 July 7.
40. Palumbo SA, Adamson KM, Krishnamurthy S, Manoharan S, Beiler D, Seiwel A, et al. Assessment of probable opioid use disorder using electronic health record documentation. *JAMA Netw Open*. 2020;3(9):e2015909. <https://doi.org/10.1001/jamanetworkopen.2020.15909> PMID: [32886123](#)